

Case File — “The Rusticle”

Client: Boğaziçi University Marine Heritage Microbiology Project, İstanbul

Sample provenance

A scraping of “rusticle” — the iron-rich, dripping, biofilm-like microbial structure that drapes the corroding steel hull of a famous early-20th-century ocean liner — was retrieved during a recent submersible dive over the wreck. From the scraping, a single moderately halophilic bacterial strain was successfully isolated and cultured on saline marine medium. Our project has sequenced its genome and would like an annotation and identification.

The brief

Whatever this organism is, it makes its home on submerged steel in cold, saline, nutrient-poor open ocean. The client wants to know:

1. **What organism is this most likely from?**
2. **What does its protein content tell you about its lifestyle?** Are there hints of how it copes with salinity, of biofilm formation, of growth in an oligotrophic marine environment, or of the iron chemistry of its substrate?
3. **Is there anything genuinely unusual or interesting in what you find?**

Materials provided

mystery_orfs.fasta — 12 predicted protein-coding ORFs from this sample’s already-assembled genome. Headers are opaque IDs (mORF_01 ... mORF_12); you will not find taxon information in the FASTA.